Hypothesis Testing for the Population Mean of a Normal Distribution

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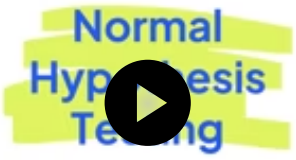

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Normal hypothesis testing

How do I do a hypothesis test with a normal distribution?

- The **population parameter** being tested will be the **population mean,** in a **normally distributed random variable** $N(\mu,\sigma^2)$
  - The population mean is tested by looking at the mean of a sample taken from the population
  - The sample mean is denoted $\bar{x}$
  - For a random variable $X \sim N(\mu,\sigma^2)$ the distribution of the sample mean would be $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
- A hypothesis test is used when the value of the assumed population mean is questioned
- The **null hypothesis**, H_0 and **alternative hypothesis**, H_1 will always be given in terms of μ
  - Make sure you clearly define μ before writing the hypotheses, if it has not been defined in the question
  - The null hypothesis will always be $H_0 : \mu = \ldots$
  - The alternative hypothesis will depend on if it is a **one-tailed** or **two-tailed** test
  - A one-tailed test would test to see if the **value of μ** has **either increased or decreased**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

changed
    - The **alternative hypothesis, H_1 will be $H_1: \mu \neq ..$**
- To carry out a hypothesis test with the normal distribution, the **test statistic** will be the **sample mean, $\overline{X}$**
    - Remember that the **variance** of the sample mean distribution will be the variance of the population distribution **divided by *n***
    - the **mean** of the sample mean distribution will be the same as the mean of the population distribution

- The normal distribution will be used to **calculate the probability** of the observed value of the test statistic taking the observed value **or a more extreme value**
- The hypothesis test can be carried out by
    - either calculating the probability of the test statistic taking the observed or a more extreme value (**p – value**) and comparing this with the significance level
    - or by finding the critical region and seeing whether the observed value of the test statistic lies within it
        - Finding the critical region can be more useful for considering more than one observed value or for further testing

How do I find the critical value?

- The **critical value(s)** will be the boundary of the **critical region**
    - The probability of the observed value being within the critical region, given a true null hypothesis will be the same as the **significance level**

- For an $\alpha\%$ significance level:
    - In a one-tailed test the critical region will consist of α % in the tail that is being tested for
    - In a two-tailed test the critical region will consist of $\frac{\alpha}{2}\%$ in each tail

- To find the critical value(s) find the distribution of the sample means, assuming H_0 is true, and use the inverse normal function on your calculator
- For a **two-tailed test** you will need to find **both critical values,** one at each end of the distribution

How can I use the standard normal distribution Z to perform a hypothesis test?

- You could use the standard normal distribution:
    - Find the z-value for your sample mean using $z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$
    - Find the critical value(s) for the Z distribution using the percentage points table
    - If the z-value is further away from 0 than the critical value then reject H_0

- You could use the standard normal distribution as an alternative method for finding the critical value(s) is to use the standard normal distribution:

- Step 2. Use the coding $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$ to standardise to Z
  - Step 3. Use percentage points table to find the z - value for which the probability of Z being equal to or more extreme than the value is equal to the significance level
  - Step 4. Equate this value to your expression found in step 2
  - Step 5. Solve to find the corresponding value of $\bar{x}$
- If using this method for a two-tailed test be aware of the following:
  - The symmetry of the normal distribution means that the z - values will have the same absolute value
  - You can solve the equation for both the positive and negative z – value to find the two critical values
    - Check that the two critical values are the same distance from the mean

How do I lay out the hypothesis test?

- Following these steps will help when carrying out a hypothesis test for the mean of a normal distribution:
- **Step 1.** Define the distribution of the **population mean** usually $X \sim N(\mu, \sigma^2)$
- **Step 2.** Write the null and alternative hypotheses clearly using the form

$$H_0 : \mu = \ldots$$

$$H_1 : \mu \ldots \ldots$$

- **Step 3.** Assuming the null hypothesis to be true, define the **test statistic,** usually $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$
- **Step 4.** Calculate either the **critical value(s)** or the **p – value** (probability of the observed value) for the test
- **Step 5.** Compare the observed value of the test statistic with the critical value(s) or the p - value with the significance level
- **Step 6.** Decide whether there is enough evidence to reject H_0 or whether it has to be accepted
- **Step 7.** Write a conclusion in context
- Alternatively, if you have used the standard normal distribution method then in steps 4 and 5 you could compare the z – value corresponding to the observed value with the z – value corresponding to the critical value

Worked Example

The time, X minutes, that it takes Amelea to complete a 1000–piece puzzle can be modelled using $X \sim \mathrm{N}(204, 81)$. Amelea gets prescribed a new pair of glasses and claims that the time it takes her to complete a 1000–piece puzzle has decreased. Wearing her new glasses, Amelea completes 12 separate 1000–piece puzzle and calculates her mean time on these puzzles to be 201 minutes. Use these 12 puzzles as a

Answer:

Examiner Tips and Tricks

- Use a diagram to help, especially if looking for the critical value and comparing this with an observed value of a test statistic

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